ZOO 1302
Vertebrate Diversity

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Subphylum - Vertebrata

Fish

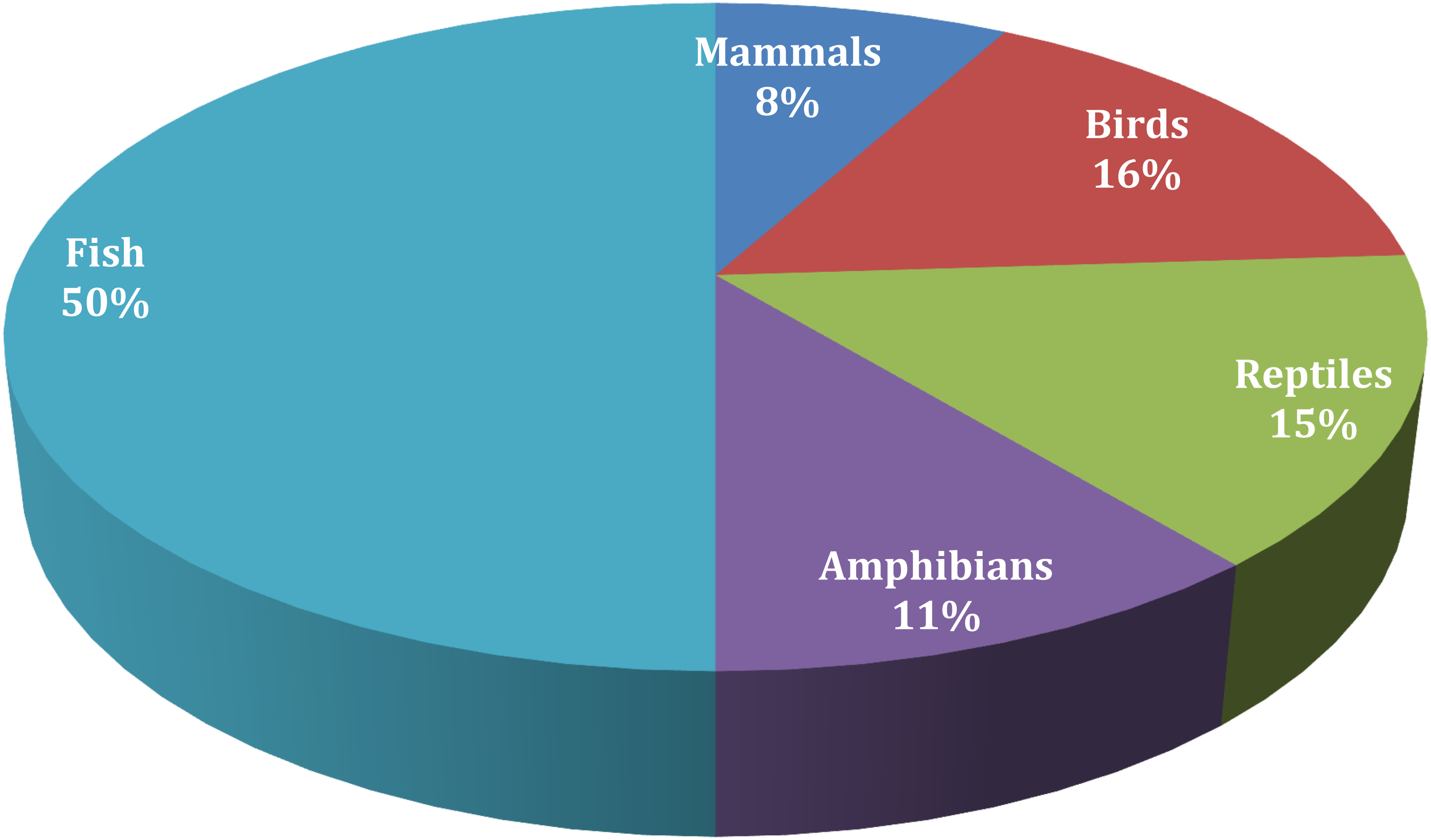
What is a fish ???

- an aquatic vertebrate
- have gills
- limbs are present in the form of fins
- skin covered with scales of dermal origin

- Of the 66,178 species of vertebrates, approximately 5,513 are mammals, 10,425 are birds, 10,038 are reptiles, 7,302 are amphibians, and 32,900 are fishes.

- Although these numbers change as new species are discovered, new fishes are found more often than other new vertebrates are found.

- As scientists make new discoveries, it is expected that the number of species of fishes will outnumber other vertebrates even more.



Relative percentage of vertebrates.
50 percent of living vertebrates are fishes.

Classification of fish

- Fishes are most ancient and most diverse of the monophyletic group and they belong to 5 classes.

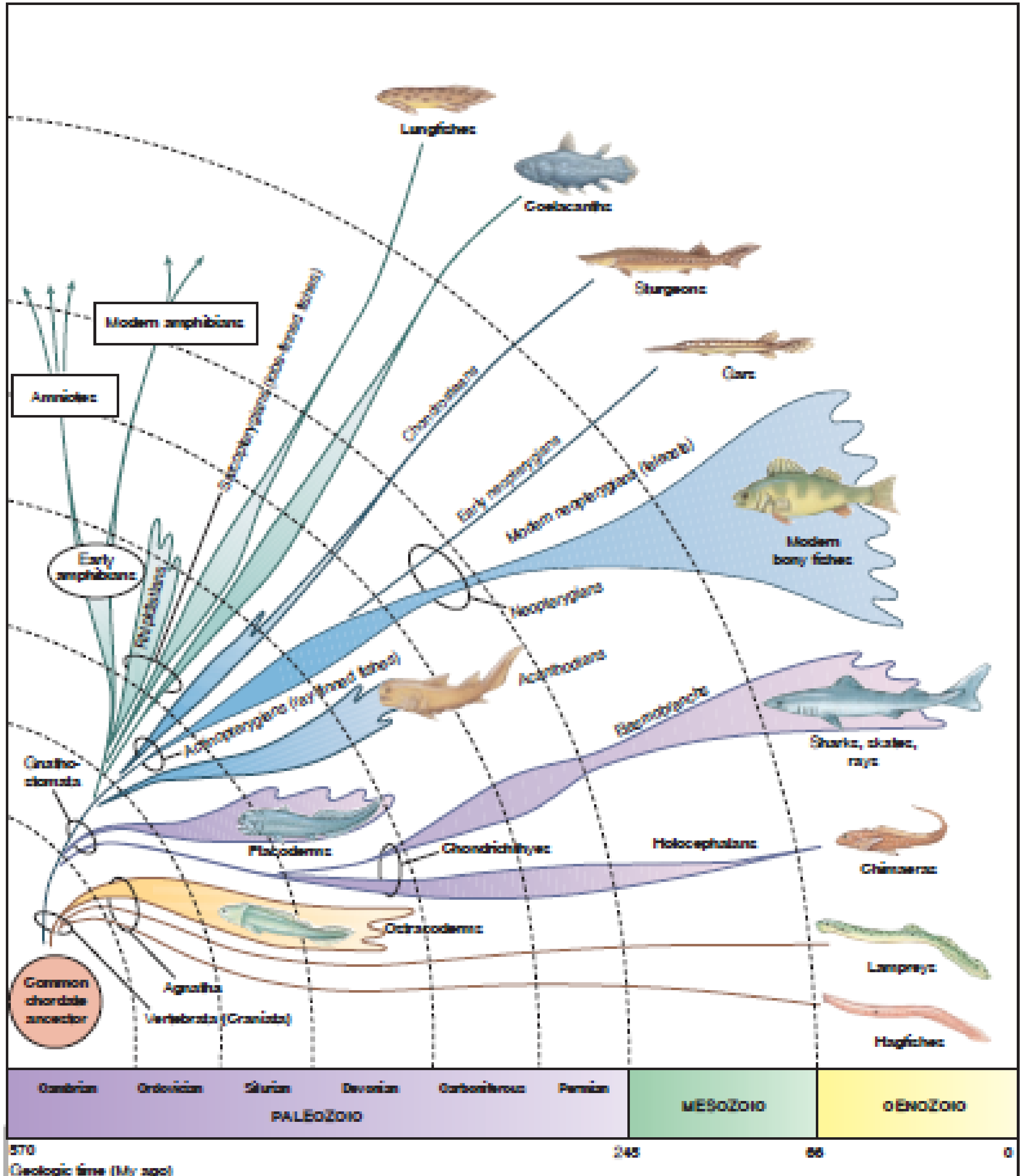
- **Phylum - Chordata**
- **Subphylum - Vertebrata**
- **Superclass - Agnatha**

1. Class - Myxini
2. Class – Cephalaspidomorphi (Petromyzontida)

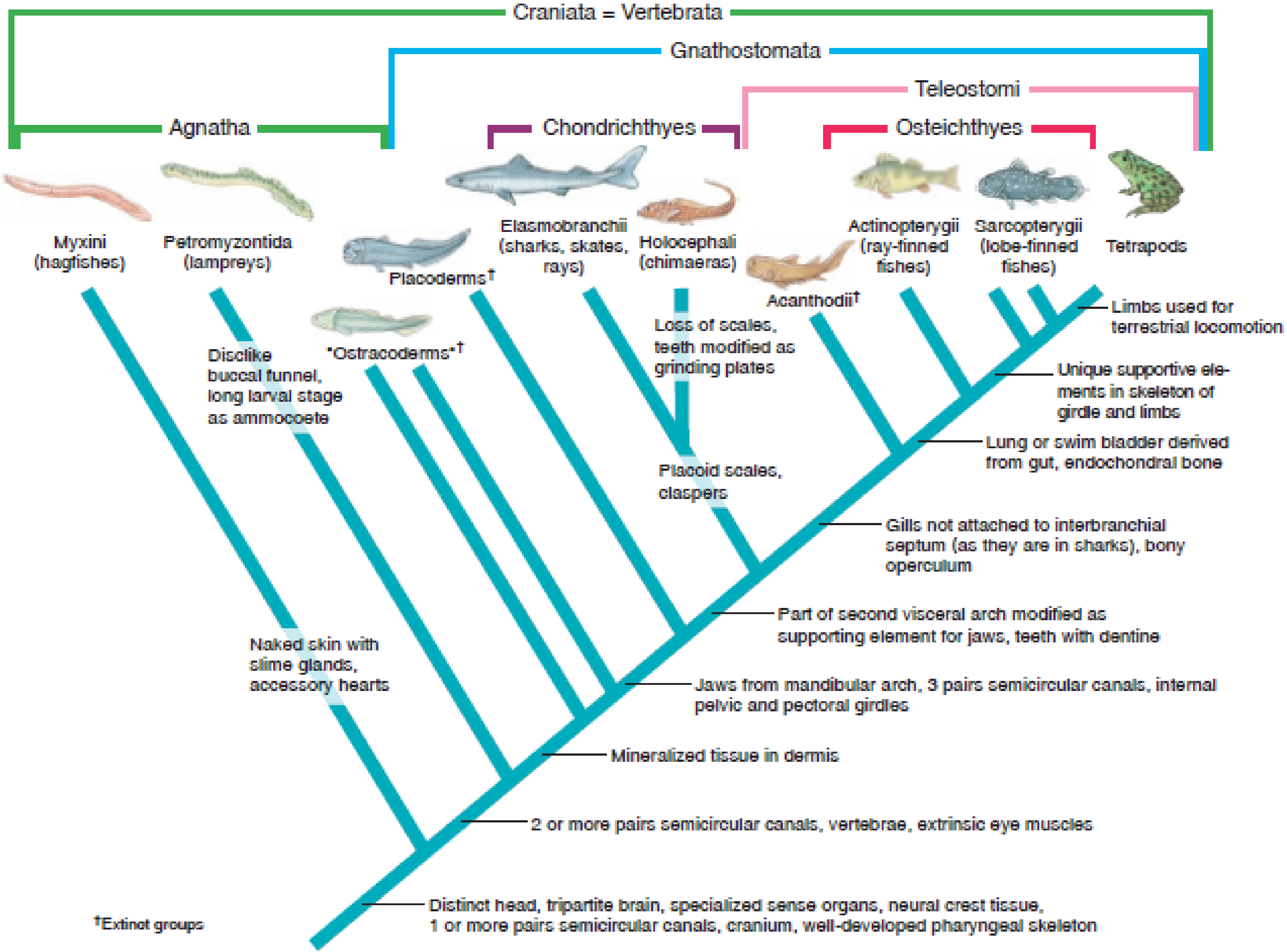
Phylum – Chordata
Subphylum– Vertebrata
Superclass - Agnatha

3. Class – Chondrichthyes
4. Class - Actinopterygii
5. Class - Sarcopterigii

Phylum – Chordata
Subphylum – Vertebrata
Superclass - Gnathostomata



Lungfishes
Coelacanths
Sturgeons
Gars
Modern amphibians
Amniotes
Sarcopterygians (lobe-finned fishes)
Chondrosteans
Early neopterygians
Modern neopterygians (teleosts)
Modern bony fishes
Early amphibians
Rhipidistians
Neopterygians
Actinopterygians (ray-finned fishes)
Acanthodians
Elasmobranchs
Sharks, skates, rays
Gnatho-stomata
Placoderms
Chondrichthyes
Holocephalans
Chimaeras
Ostracoderms
Lampreys
Agnatha
Vertebrata (Craniata)
Common chordate ancestor
Hagfishes
Cambrian
Ordovician
Silurian
Devonian
Carboniferous
Permian
PALEOZOIC
MESOZOIC
CENOZOIC
570
245
66
0
Geologic time (My ago)


Craniata = Vertebrata
Gnathostomata
Teleostomi
Agnatha
Chondrichthyes
Osteichthyes
Myxini (hagfishes)
Petromyzontida (lampreys)
Placoderms†
"Ostracoderms"†
Elasmobranchii (sharks, skates, rays)
Holocephali (chimaeras)
Acanthodii†
Actinopterygii (ray-finned fishes)
Sarcopterygii (lobe-finned fishes)
Tetrapods
Disclike buccal funnel, long larval stage as ammocoete
Naked skin with slime glands, accessory hearts
Loss of scales, teeth modified as grinding plates
Placoid scales, claspers
Limbs used for terrestrial locomotion
Unique supportive elements in skeleton of girdle and limbs
Lung or swim bladder derived from gut, endochondral bone
Gills not attached to interbranchial septum (as they are in sharks), bony operculum
Part of second visceral arch modified as supporting element for jaws, teeth with dentine
Jaws from mandibular arch, 3 pairs semicircular canals, internal pelvic and pectoral girdles
Mineralized tissue in dermis
2 or more pairs semicircular canals, vertebrae, extrinsic eye muscles
Distinct head, tripartite brain, specialized sense organs, neural crest tissue, 1 or more pairs semicircular canals, cranium, well-developed pharyngeal skeleton
†Extinct groups

Lesson 5

Superclass – Agnatha

Jawless Fish

Superclass – Agnatha (Jawless fish)

- Approximately 84 species living today
- Two classes;
  - a) Class - Myxini (Hagfish)
  - b) Class - Cephalaspidomorphi (Petromyzontida) (Lampreys)
- Lack ………….
  - ✓ jaws
  - ✓ internal ossification
  - ✓ paired fins
  - ✓ scales

- The ancestor is uncertain.
- Superficially they look much alike.
- Have Pore-like gill openings.
- Eel-like body form.
- But are so different from each other and assigned to separate classes by ichthyologists.
- Hagfishes have no vertebrae and lampreys have only rudimentary vertebrae.
- But they have a cranium and many other vertebrate homologies – included to subphylum Vertebrata.

Class - Myxini (Hagfishes)

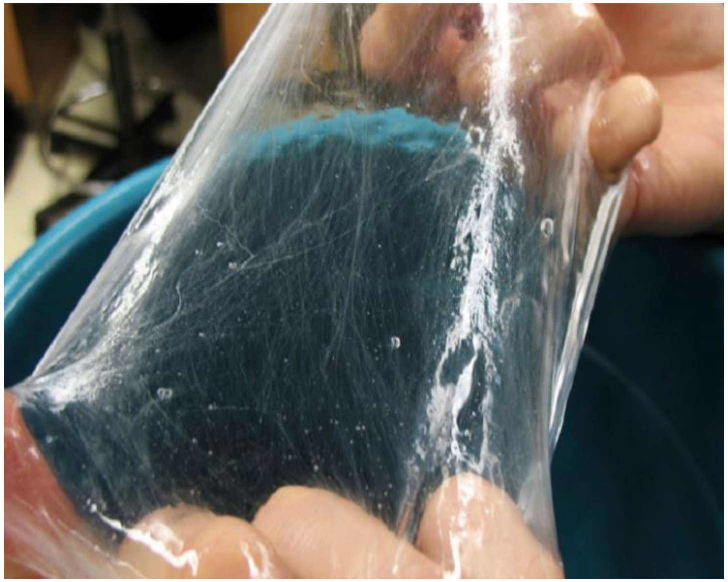
Class - Myxini (Hagfish)

- Entirely marine, about 70 species
- Body slender, eel-like, rounded, with naked skin containing slime glands
- Mouth is surrounded by barbels
- No paired appendages
- No dorsal fin
- Almost completely blind
- Fibrous and cartilaginous skeleton
- Notochord persistent


barbels
no eyes
mucilaginous pores
50 mm
Fisheries and Oceans Canada C. Nozères

- Gills – 5 to 16 pairs of gills with a variable number of gill openings
- Heart - with sinus venosus, atrium and ventricle; accessory hearts, aortic arches in gill region
- Kidney - Segmented mesonephric kidney - marine body fluid isosmotic with seawater
- Nervous system - Dorsal nerve cord with differentiated brain; no cerebellum; 10 pairs of cranial nerves; dorsal and ventral nerve roots united
- Sensory system - Sense organs of taste, smell and hearing; eyes degenerate
- Reproductive system - Sexes separate; External fertilization; Large yolky eggs; No larval stage

- Hagfishes are renowned for their ability to generate enormous quantities of slime.
- If disturbed or roughly handled, a hagfish exudes a milky fluid from special glands positioned along its body.
- On contact with seawater, the fluid forms a slime so slippery that the animal is almost impossible to grasp.



Feeding

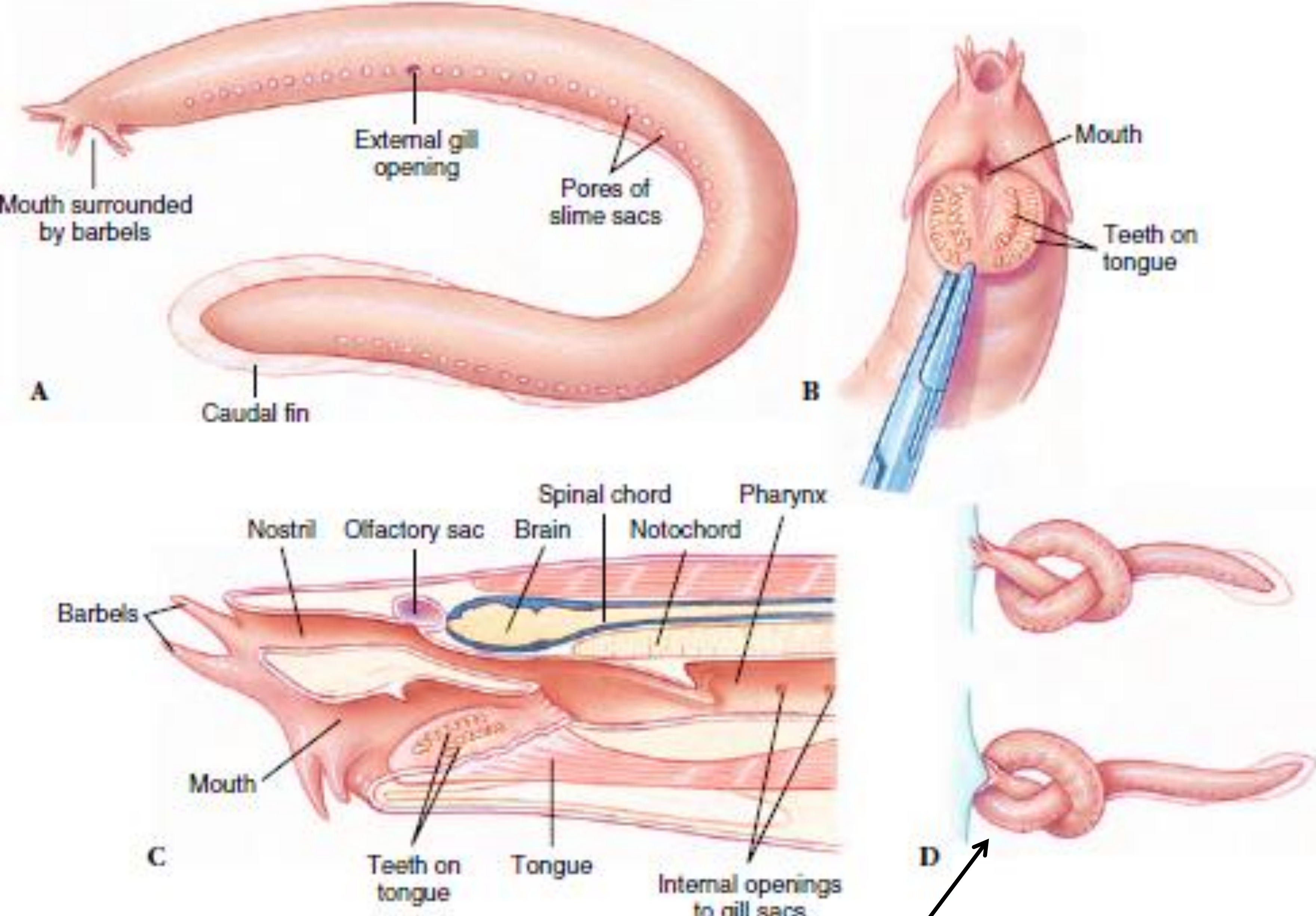
- Hag fishes feeds on annelids, molluscs, crustaceans, and dead or dying fishes.

- They are not parasitic like lampreys but are scavengers and predators.

- Although almost completely blind, the hagfish is quickly attracted to food, especially dead or dying fishes, by its keenly developed senses of smell and touch.

- The hagfish enters a dead or dying animal through an orifice or by digging inside.

- They have biting mouth with two rows of eversible teeth.
- Using two toothed, keratinized plates on the tongue that fold together in a pincerlike action, the hagfish rasps away bits of flesh from its prey.
- For extra leverage, the hagfish often ties a knot in its tail, then passes it forward along the body until it is pressed securely against the side of its prey.
- Digestive system - without stomach
- No spiral valve or cilia in intestinal tract.



Hagfish knotting – for obtain leverage to tear flesh from prey


Closed mouth


Open mouth
side view


Open mouth
front view

Class – Cephalaspidomorphi (Petromyzontida) (Lampreys)

Class – Cephalaspidomorphi (Petromyzontida) (Lampreys)

- All the lampreys of the Northern Hemisphere belong to the family Petromyzontidae – entirely marine
- The group name refers to the lamprey's habit of grasping a stone with its mouth to hold position in a current (Gr. *Petros – stone* , *myzo* -sucking).
- There are 22 species of lampreys in North America.
- About half of these belong to the nonparasitic brook type and the others are parasitic.

- *Petromyzon marinus* - destructive, parasitic marine lamprey is found on both sides of the Atlantic Ocean (in America and Europe) - attain a length of 1 m.

- *Lampetra* - (L. lambo, to lick or lap up) distributed in North America and Eurasia – size 15 to 60 cm in length.

  eg. *Lampetra tridentatus* – chief marine form in west coast of North America

- *Ichthyomyzon (*Gr. ichthyos, fish, myzon, sucking) - includes three parasitic and three nonparasitic species and restricted to eastern North America.

- All lampreys ascend freshwater streams to breed.

Characteristics - Eg. *Petromyzon marinus*

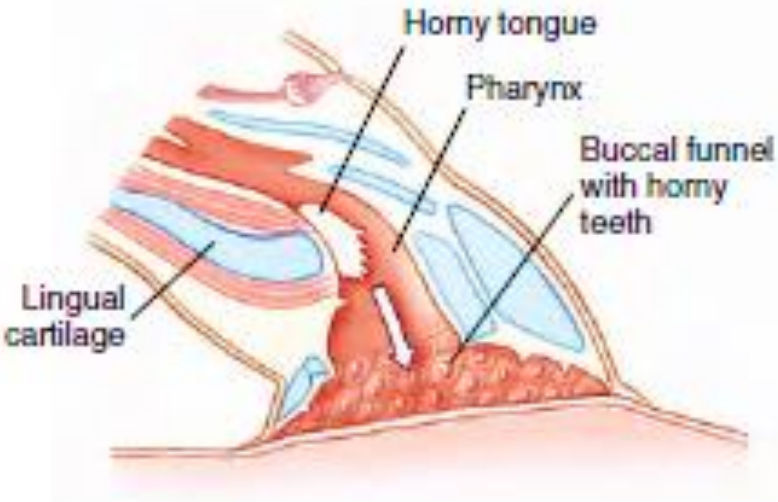
- Body slender, eel-like, rounded, with naked skin containing slime glands
- No paired appendages, one or two median fins
- Seven pairs of gills each with external gill opening
- Fibrous and cartilaginous skeleton, notochord persistent

- Heart - with sinus venosus, atrium and ventricle, aortic arches in gill region
- Kidney - Opisthonephric kidney – a kidney that develops from the middle and posterior portions of nephrogenic region of vertebrates
- Nervous system - Dorsal nerve cord with differentiated brain, small cerebellum present; 10 pairs of cranial nerves; separated dorsal; and ventral nerve roots
- Sensory system - sense organs of taste, smell, hearing; eyes well developed in adults; two pairs of semicircular canals

Feeding

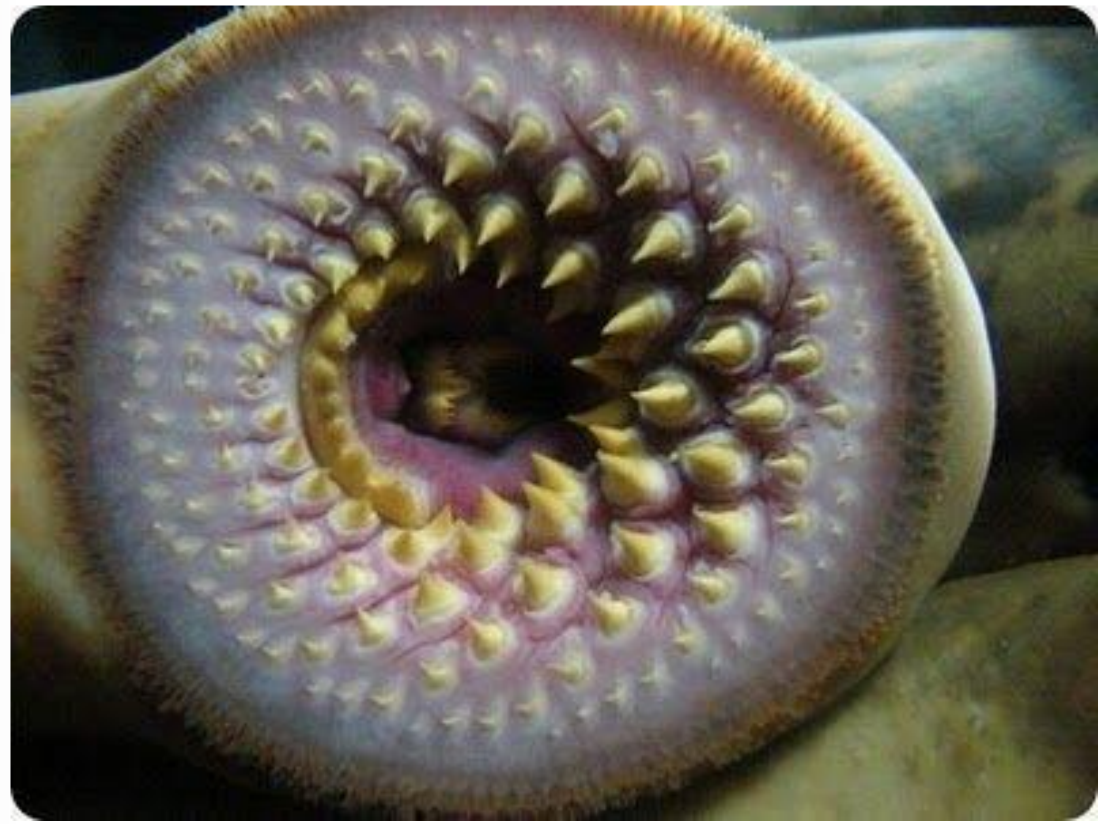
- Sucker like oral disc and tongue with well developed keratinized teeth.

- During feeding they attach themselves by their sucker like mouth to a fish.
- With their sharp keratinized teeth, rasp away the flesh and suck out body fluids.
- To promote the flow of blood, the lamprey injects an anticoagulant into the wound.
- When gorged, the lamprey releases its hold but leaves the fish with a large, gaping wound that is sometimes fatal.
- Digestive system - without stomach; intestine with spiral fold



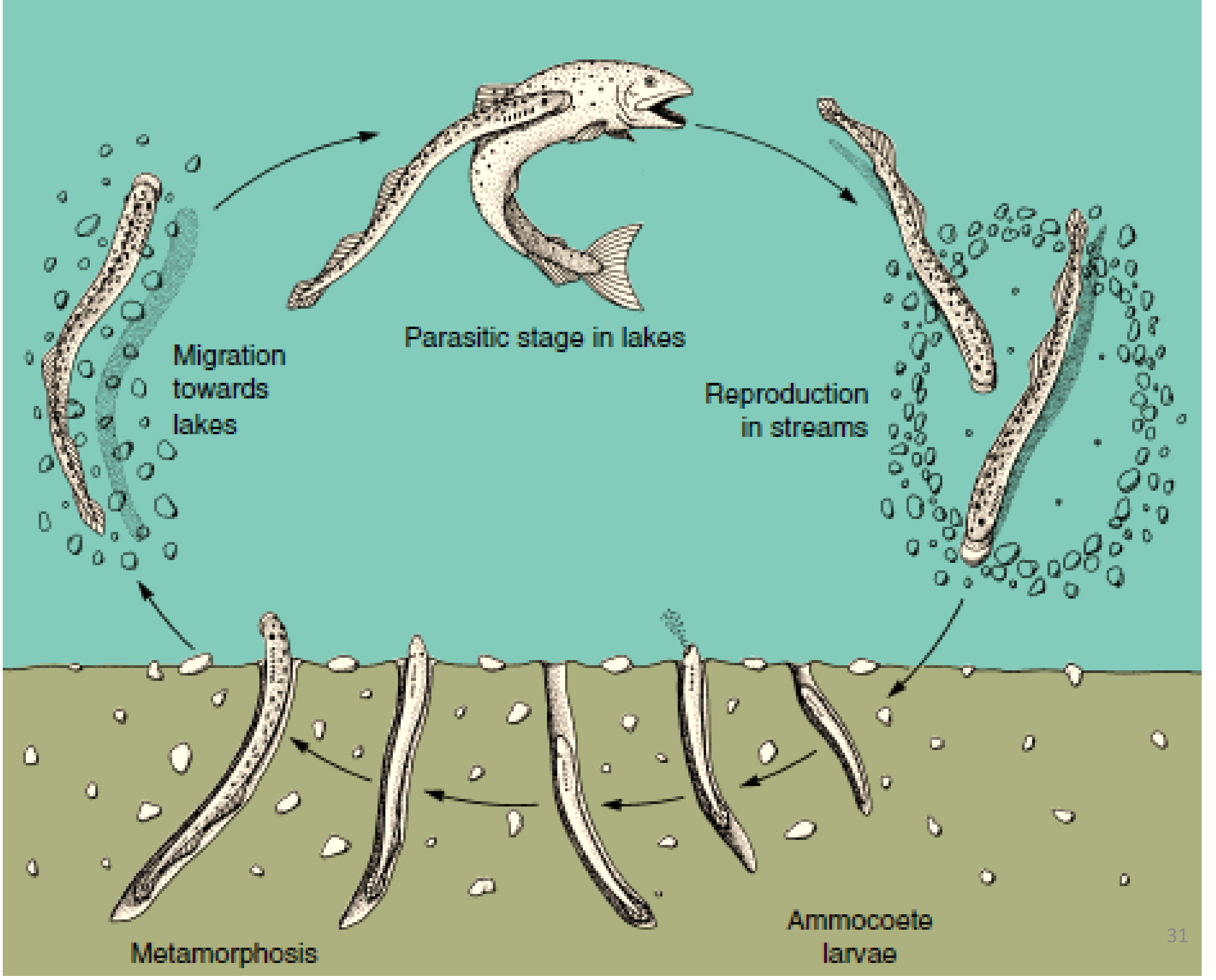
Attachment to fish with horny teeth and suction

Tongue protruded for rasping flesh



Reproduction

- Sexes separate; single gonad without duct
- Anadromous (migrate from marine to freshwater); body fluids osmotically regulated
- Males - begin to building a nest and are jointed later by females.
- Using their oral discs to lift stones and pebbles and vigorous body vibrations to sweep away light debris, they form an oval depression.
- At spawning - with the female attached to a rock to maintain her position over the nest, the male attaches to the dorsal side of her head.


Parasitic stage in lakes
Migration towards lakes
Reproduction in streams
Metamorphosis
Ammocoete larvae

- As eggs are shed into the nest, they are fertilized by the male - external fertilization.

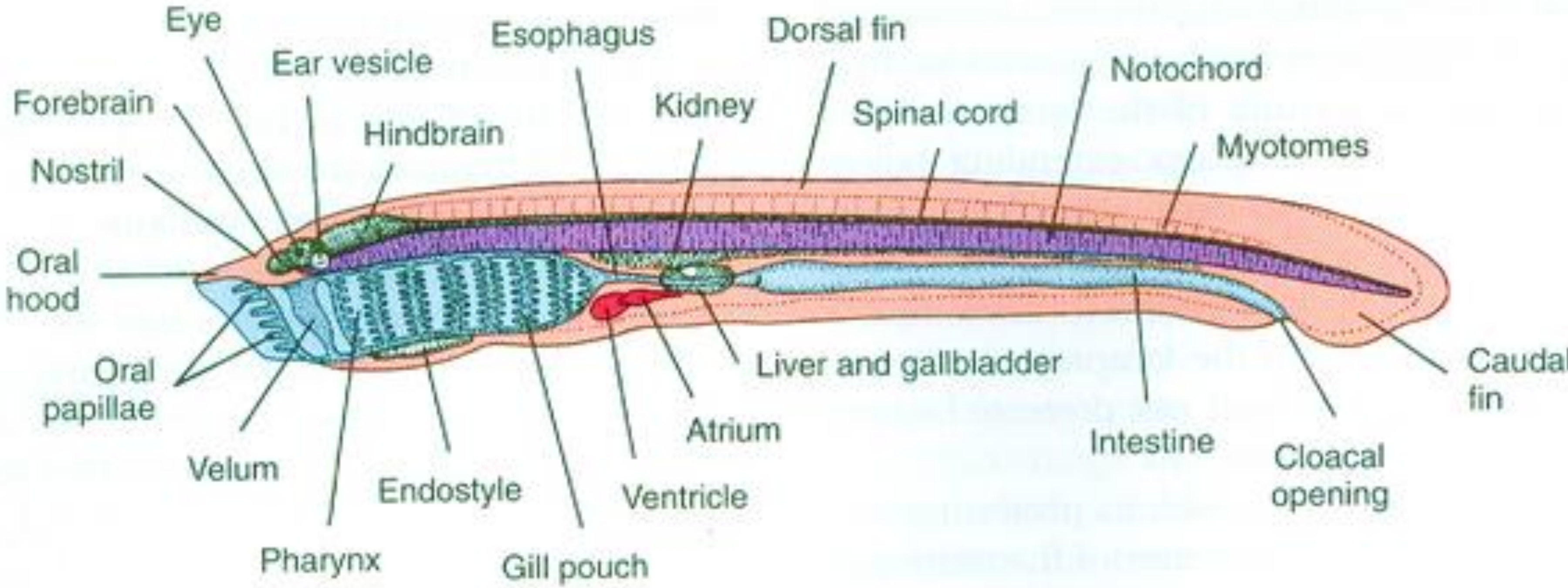
- The sticky eggs adhere to pebbles in the nest and quickly become covered with sand.

- Adults die soon after spawning.

- Eggs hatch in about 2 weeks

- Small long larval stage - ammocoete larva

- After absorbing the remainder of their yolk supply, young ammocoetes - 7 mm long.


Eye
Esophagus
Dorsal fin
Ear vesicle
Notochord
Forebrain
Kidney
Spinal cord
Hindbrain
Myotomes
Nostril
Oral hood
Oral papillae
Velum
Pharynx
Endostyle
Gill pouch
Ventricle
Atrium
Liver and gallbladder
Intestine
Cloacal opening
Caudal fin

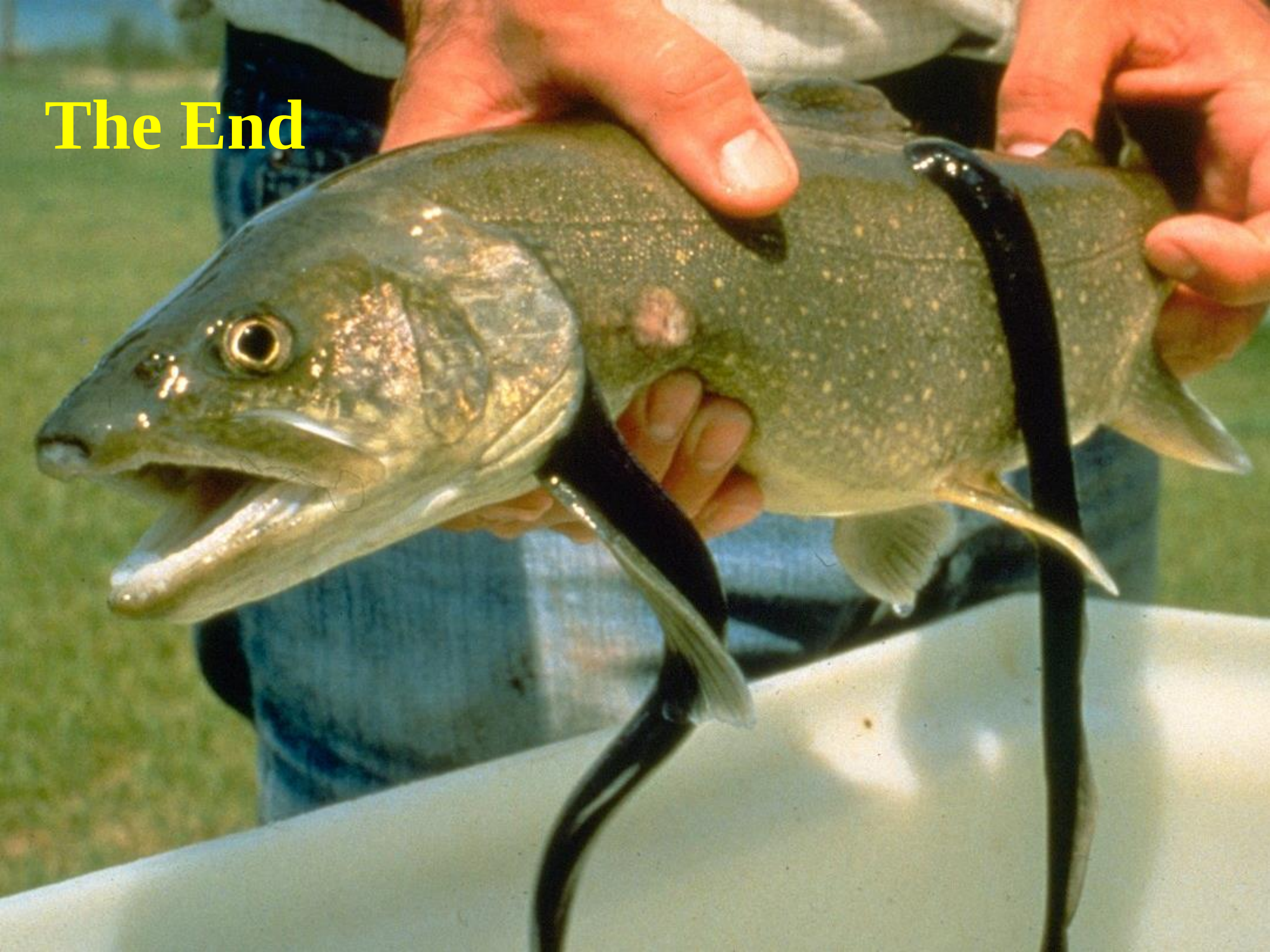
- Larva bears a remarkable resemblance to amphioxus
- Hence it is considered a chordate archetype.
- It leave the nest gravel and drift downstream to burrow in a suitable sandy, low-current area.
- Larvae live as suspension feeders
- Larva grows slowly for 3 to 7 or more years, then rapidly metamorphose into adults.

- Changes during metamorphosis
  - eruption of eyes,
  - replacement of the hood by the oral disc with keratinized teeth,
  - enlargement of fins,
  - maturation of gonads, and
  - modification of the gill openings.

- Parasitic lampreys either migrate to the sea if marine, or remain in freshwater.
- Parasitic freshwater adults live 1 to 2 years before spawning and then die.
- Anadromous forms live 2 to 3 years.
- Nonparasitic lampreys
  - do not feed after emerging as adults
  - their digestive tract degenerates to a nonfunctional strand of tissue.
  - within a few months they spawn and die.

Compare

- Hagfish and lamprey
- Ammocoete larva and amphioxus


The End